Ghosh's monic sextic equation has roots expressible in radicals

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Abstract

In this paper, I describe the roots of my monic sextic equation, which are expressible in radicals. The paper ends with "The End"

Introduction

In a previous paper, I've described one of the roots of the quartic equation and how that can be used to solve the general quartic equation in radicals. In a previous paper, I've described my monic sextic. In this paper, I describe the roots of my monic sextic equation, which are expressible in radicals.

Ghosh's monic sextic equation has roots expressible in radicals

When

$$Q \neq 0$$

by the right hand side of Ghosh's monic sextic identity,

Ghosh's monic sextic equation

$$\left(x^4 + Px^3 + \left(b - aP + P^2 - Q\right)x^2 + \left(\frac{eQ + fP - af}{Q^2}\right)x + \frac{f}{Q}\right)\left(x^2 + (a-P)x + Q\right) = 0$$

can be written as the product of a quartic equation and a quadratic equation, both of which can be solved in radicals.

The End